

CSCI 242: Algorithms and Data Structures

PROJECT 1 (100 points)

Instructions:

Choose a sorting algorithm from the list below (check in the discussion board that the sorting algorithm you decided on is not already chosen by another student or team) and put your choice in the **discussion board#1** (Assignment Sorting algorithms) on Blackboard. The goal of this project is to do a research report on the sorting algorithm (chosen by your team).

The points you might want to discuss in your report are, but not limited to:

- The sorting algorithm and its implementation (by providing the algorithm and pseudocode)
- Applications of the sorting algorithm
- Time complexity of the sorting algorithm (provide the asymptotic notations as well)
- Three cases of the algorithm (best, average, and worst)
- Comparison of the performance of the sorting algorithm with other sorting algorithms

Objective:

1. Write a research report on the sorting algorithm you chose, focusing on the goals mentioned above (Between 3 to 5 pages). The output is a .pdf file.
2. A ppt presentation and an oral presentation of the research report. The presentation should be approximately 15 minutes (If working as a team, split the presentation between team members. If 3 members in the team, each student represents for 5 minutes). The quality of both the ppt and the oral presentation will be graded, besides the report. The output is a .ppt file.
3. Submit the pdf report and the ppt presentation as a zipped folder via Blackboard. Include your name (team members name) inside each document.

List of sorting algorithms:

- Selection Sort
- Bubble Sort
- Recursive Bubble Sort
- Quick Sort
- Insertion Sort
- Bucket Sort
- Radix Sort
- Count Sort
- Heap Sort
- Tree Sort
- ShellSort
- TimSort